TIMEMIXER++: A GENERAL TIME SERIES PATTERN MACHINE FOR UNIVERSAL PREDICTIVE ANALYSIS

Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenze Lin, Shengtong Ju, Zhixuan Chu, Ming Jin

Griffith University, The Hong Kong University of Science and Technology (Guangzhou), Massachusetts Institute of Technology, Zhejiang University, The State Key Laboratory of Blockchain and Data Security, Hangzhou High-Tech Zone (Binjiang) Institute of Blockchain and Data Security

May 24, 2026

Why Time Series Matter

- Time series power decisions in energy, traffic, finance, climate, industry, and healthcare.
- Core tasks: forecasting, classification, anomaly detection, imputation, few/zero-shot transfer.
- Real-world sequences are multi-scale and multi-periodic with noise, missingness, and long-range dependencies.

Goal: One Model for Many

- Existing models excel on one task but struggle to generalize across tasks and domains.
- Challenges: overlapping periodicities, season–trend entanglement, long-range dependencies.
- Need a universal Time Series Pattern Machine that adapts to diverse tasks without task-specific redesign.

Idea and Contributions

- Convert sequences into multi-resolution time images to expose temporal–frequency structure.
- Disentangle seasonality and trend via dual-axis attention; hierarchically fuse across scales and periods.
- Deliver a single backbone with task-adaptive heads achieving SOTA across 8 tasks and 30+ datasets.

Architecture Overview

- Pipeline: MRTI → TID → MCM → MRM, repeated in MixerBlocks with residual connections.
- Multi-scale inputs; per-scale, per-period processing; task-specific heads with ensemble across scales.
- Encoder-only, efficient design with strong accuracy–efficiency trade-offs.

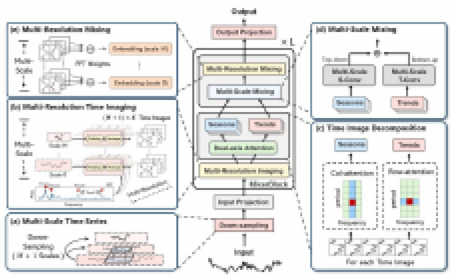

Figure: TIMEMIXER++：序列投影+堆叠MixerBlocks，生成多分辨率时间图，双轴注意力解耦季节/趋势并多尺度混合

Input: Scales and Channels

- Downsample input into M+1 scales via strided convs; embed to shared d_model.
- Coarsest scale applies variate-wise self-attention to capture inter-variable dependencies.
- Provides global context early to stabilize downstream multi-scale processing.

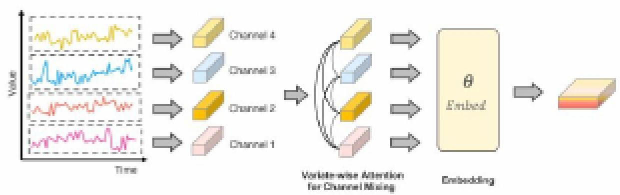

Figure: 输入投影中的通道混合与嵌入函数：变量级自注意在最粗尺度建模跨变量依赖，随后投影至嵌入空间

Stage 1: Time Imaging

- FFT at coarsest scale to select top-K dominant periods per sequence.
- Reshape each scale into K period-aligned 2D time images (rows: period; cols: segments).
- Preserves temporal order while exposing periodic/frequency structure for 2D operations.

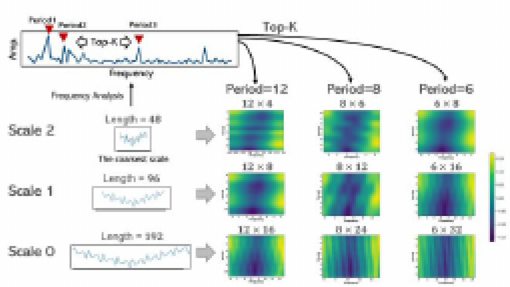

Figure: 多分辨率时域成像：基于前三频率与三尺度生成图像

Stage 2: Dual-Axis Decomposition

- Column-axis attention models within-period seasonality (across columns).
- Row-axis attention models across-period trend (across rows).
- Shared 2D convs produce Q/K/V; outputs are disentangled seasonal and trend images.

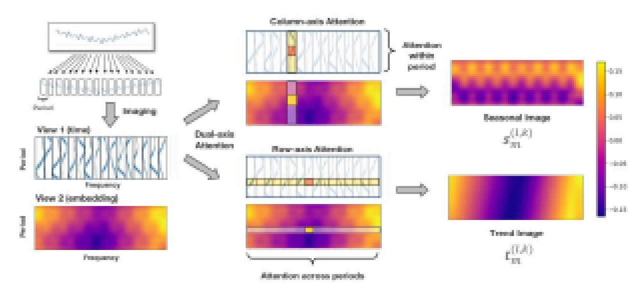

Figure: 时间图像分解：基于识别周期构建时间图像，并用双轴注意力提取季节性与趋势

Stage 3: Multi-Scale Mixing

- Seasonal fusion bottom-up (fine → coarse) via residual 2D convolutions for long motifs.
- Trend fusion top-down (coarse → fine) via transposed 2D convolutions for local detail.
- Produces scale-aware season–trend images per period.

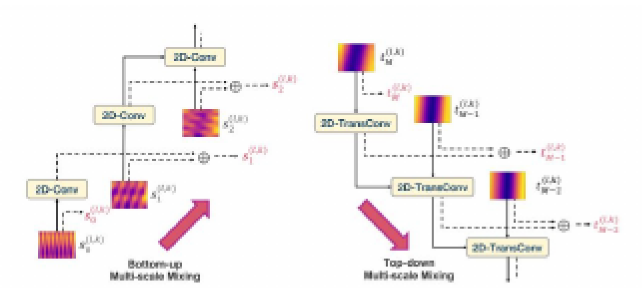

Figure: 多尺度层次混合季节/趋势图像：输出累积先验；自底向上2D卷积，自顶向下反卷积适配尺度。

Stage 4: Multi-Resolution Mixing

- For each scale, fuse K period-specific sequences using softmax-normalized FFT amplitudes.
- Amplitude-aware weights prioritize dominant periodicities while retaining others.
- Outputs one robust representation per scale for ensembling.

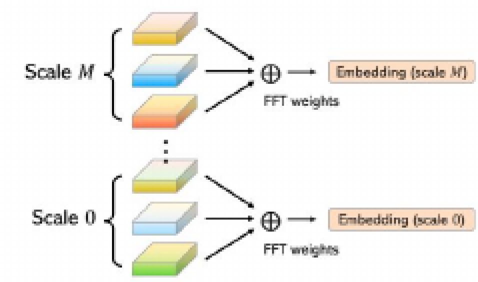

Figure: Multi-resolution: S–T mix; fuse K period reps → M scales

Outputs and Training

- Multi-head predictors (one per scale); final output via scale-wise ensemble.
- Encoder-only backbone with hierarchical conv mixing and limited attention usage.
- Scalable hyperparameters (M, L, K) tuned by sequence length and task.
- Favorable compute–accuracy trade-offs; encoder-only with limited attention and 2D convs scales well for long inputs and high-dimensional series.

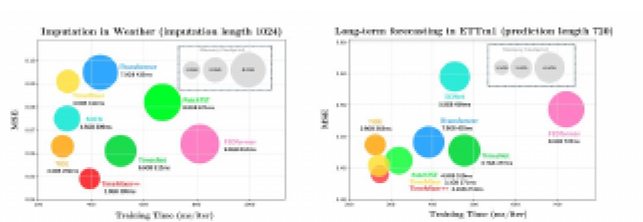

Figure: Model efficiency comparison under imputation and long-term forecasting.

Setup: Data and Baselines

- 8 task types over 30+ datasets: long/short-term forecasting, imputation, classification, anomaly, few/zero-shot.
- Long-term: ETT*, Weather, Solar, Electricity, Traffic; Short-term: M4 (uni), PEMS (multi).
- Comparisons against 27 baselines: Transformers, CNN/TCN, MLP, frequency-aware, and classical series models (e.g., iTransformer [Zeng et al., 2023], PatchTST [Nie et al., 2023], FEDformer [Zhou et al., 2022], TimesNet [Wu et al., 2023], TimeMixer [Chen et al., 2024], TiDE [Das et al., 2023], DLinear [Zeng et al., 2023a], SCINet [Liu et al., 2022], MICN [Wang et al., 2023], Autoformer [Wu et al., 2021], Stationary [Liu et al., 2022b], Informer [Zhou et al., 2021], Reformer [Kitaev et al., 2020], Crossformer [Zhang & Yan, 2023], ETSformer [Woo et al., 2022]).

Main Results: Eight Tasks

- Consistent SOTA across forecasting, imputation, classification, anomaly detection.
- Notable gains: up to 7.3% MSE (Electricity), 30.8% MAE (PEMS), +2.59% F1 (anomaly).

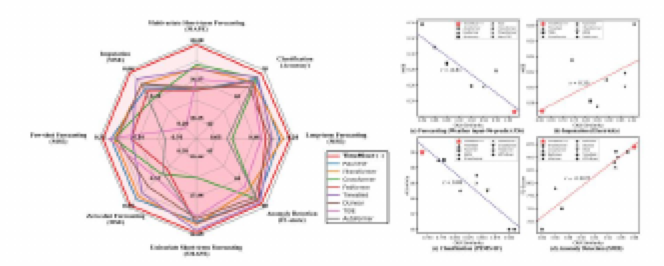

Figure: Benchmark across eight tasks; our model attains consistent SOTA across forecasting, imputation, classification, and anomaly detection.

Limitations

- Assumes quasi-stationary periods per window; rapidly shifting regimes may challenge FFT-based period selection.
- Irregular sampling and event-based streams are not directly addressed.
- Ensembling across scales adds minor inference overhead in latency-critical settings.

Long-Term Forecasting Results

- TimeMixer++ leads or ranks top-2 on Electricity, ETT avg, Solar, Traffic, Weather.
- Electricity: 7.3% MSE vs iTransformer; Solar-Energy: 6.0%/9.2% MSE/MAE over second best.
- Strong across varying dimensionality and periodic structures.

Table: Table 1: Long-term forecasting results. We average the results across 4 prediction lengths: {96, 192, 336, 720}. The best performance is highlighted in red, and the second-best is underlined. Full results can be found in Appendix H.

[illegible]	[illegible]

Short-Term Forecasting: PEMS

- Large improvements on traffic datasets (PEMS03/04/07/08) vs strong baselines.
- Average MAE/MAPE/RMSE gains: up to 19.9%/19.6%/13.5% vs iTransformer; 8.6%/4.8% vs TimeMixer.
- Better multi-node temporal coherence via scale–period mixing.

Imputation Under High Missingness

- Average MSE/MAE reductions of 25.7%/17.4% vs second-best across ETT avg, Electricity, Weather.
- Stable even with input length 1024 and masking up to 50%.
- Time-image representations help interpolate seasonal/trend structure.

Table: Table 4: Results of imputation task across six datasets. To evaluate our model performance, we randomly mask {12.5%, 25%, 37.5%, 50%} of the time points in time series of length 1024. The final results are averaged across these 4 different masking ratios.

(2024) (2023a) (2023) (2023) (2023a) (2021)	(Ours) (2023) DLinear TimesNet MICN Autoformer	(2024b) (2023)	Models TimeMixer++ TimeMixer iTransformer PatchTST Crossformer FEDformer (2022b)	TIDE
Metric			MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE MSE MAE	
ECL			ETT(Avg) 0.055 0.154 0.097 0.220 0.096 0.205 0.120 0.225 0.250 0.258 0.134 0.230 0.314 0.366 0.115 0.229 0.079 0.182 0.119 0.234 0.104 0.215 0.109 0.197 0.142 0.261 0.140 0.223 0.129 0.198 0.125 0.204 0.181 0.314 0.182 0.202 0.080 0.200 0.135 0.255 0.138 0.246 0.141 0.234 Weather 0.049 0.078 0.081 0.114 0.095 0.102 0.082 0.148 0.150 0.111 0.064 0.139 0.063 0.131 0.071 0.107 0.061 0.088 0.075 0.126 0.066 0.107	

Classification and Anomaly Results

- Classification (UEA, 10 datasets): 75.9% avg accuracy, +2.3% over TimesNet.
- Anomaly detection (SMD, SWaT, PSM, MSL, SMAP): F1=87.47%, +2.59% over TimesNet.
- Demonstrates robust pattern learning from time images beyond pure forecasting.

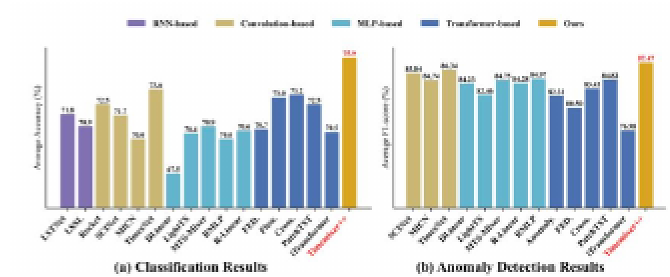

Figure: 分类与异常检测结果（跨数据集均值）。

What Time Images Learn

- Clear disentanglement of seasonal and trend components across scales.
- Multi-periodicity captured at different resolutions; patterns become spatially local.
- Helps stabilize downstream prediction and detection tasks.

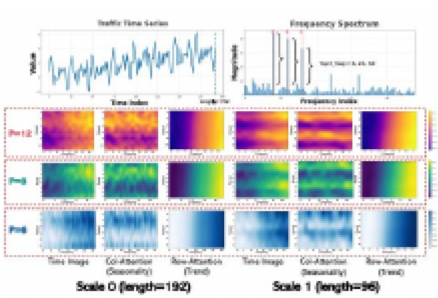

Figure: Traffic数据集的Time Image表征可视化，详见图10/11/13

Conclusion: Toward Universal TSPMs

- Introduced TimeMixer++: imaging + dual-axis attention + hierarchical mixing for robust patterns.
- Unified backbone achieves SOTA across 8 tasks on 30+ datasets with strong efficiency.
- Time images provide a practical path to disentangle and fuse multi-scale, multi-periodic structure.

Future Directions and Impact

- Scale up data/parameters to explore time series scaling laws and foundation-model paradigms.
- Extend to irregular sampling, causal inference, and decision-making pipelines.
- Enhance adaptive ensembling and uncertainty quantification for mission-critical deployments.

Baseline References

- iTransformer — Zeng, Chen, Zhang, et al., 2023, ICLR.
- PatchTST — Nie, Nguyen, Sinthong, Kalagnanam, 2023, ICLR.
- FEDformer — Zhou, Zhang, Peng, et al., 2022, ICML.
- TimesNet — Wu, Xu, Wang, et al., 2023, NeurIPS.
- TimeMixer — Chen, Xu, Zhou, et al., 2024, NeurIPS.
- TiDE — Das, Lopez, Le, et al., 2023, ICML.
- DLinear — Zeng, Chen, Zhang, 2023a, AAAI.
- SCINet — Liu, Wu, et al., 2022, AAAI.
- MICN — Wang, Chen, et al., 2023, KDD.
- Autoformer — Wu, Xu, Wang, et al., 2021, NeurIPS.
- Stationary — Liu, Zeng, et al., 2022b, NeurIPS.
- Informer — Zhou, Zhang, Peng, et al., 2021, AAAI.
- Reformer — Kitaev, Kaiser, Levskaya, 2020, ICLR.
- Crossformer — Zhang, Yan, 2023, ICLR.
- ETSformer — Woo, Dey, et al., 2022, NeurIPS.

Questions

- Thank you for your attention!
- Questions and feedback are welcome.

Acknowledgments

- We thank collaborators and the broader time series community for datasets and baselines.
- Gratitude to supporting institutions and any funding agencies involved.